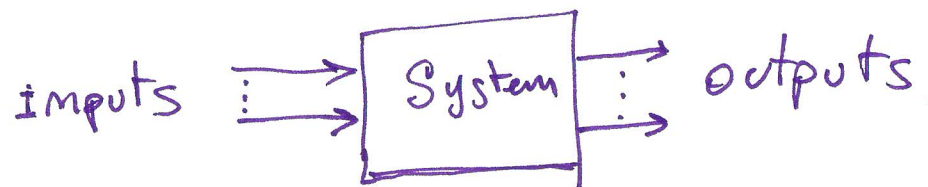


System Identification (+ Estimation)

(11)

S.I. creating models for dynamical systems.



Inputs are independent of system's evolution.

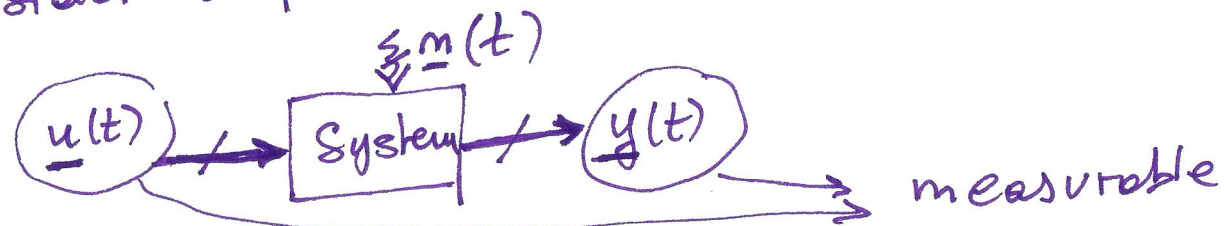
Outputs depend on $\begin{cases} \text{inputs} \\ \text{initial conditions} \end{cases}$

We can choose what are the outputs we are interested in.

Inputs are given!

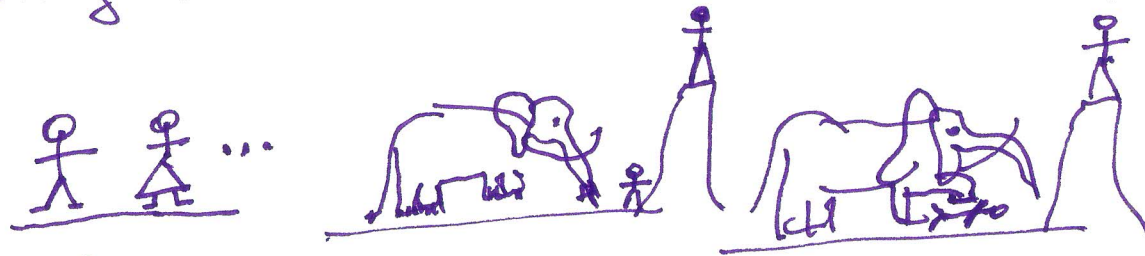
1st: understand $\begin{cases} \text{inputs} \\ \text{outputs} \end{cases}$ $\begin{cases} \text{measurable} \\ \text{non-measurable} \end{cases}$ $\begin{cases} \text{assignable} \\ \text{not in our control} \end{cases}$ $\begin{cases} \text{control} \\ \text{disturb. (noises)} \end{cases}$

We usually will consider outputs as measurable

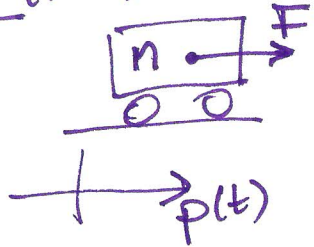


Model = ?

Anything used to describe the behaviour of a dynamical system.



Cart (no friction)



$$M \cdot a(t) = F(t)$$

$$a(t) = \frac{d}{dt} v(t) = \frac{d}{dt} \frac{d}{dt} p(t)$$

$$M \ddot{p}(t) = F(t)$$



$$\ddot{y}(t) = \frac{1}{M} u(t)$$

(differential eq.)

Models can be mathematical

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = \frac{1}{M} u(t) \\ y(t) = x_1(t) \end{cases} \quad \text{state equations.}$$

$$T_{yu}(s) = \frac{1/M}{s^2} \quad \text{Transfer Function.}$$

Why models?

(13)

- Control = give suitable inputs to obtain a satisfactory output
- Simulation = using models to see what the behaviour of the system could be after applying some control action.
- Prediction : trying to guess future behaviour of a real system. (inputs)
- Estimation: use of measurable outputs^r to compute non-measurable quantities.
- Diagnosis : detect or predict possible faults in a system.

⋮

~~Building~~ Creating a model = modelling

System Identification $\leftrightarrow$ (Parameter) Identification of a model.

SS
modelling

SS
Estimation (computation) of parameters

Different types of models.

- graphical (diagrams, drawings, ...)
- mathematical
 - code (programming code)
 - physical (Toy car, scale models).
 - real system itself.

models can be more/less accurate
more/less complex } depending on the application.